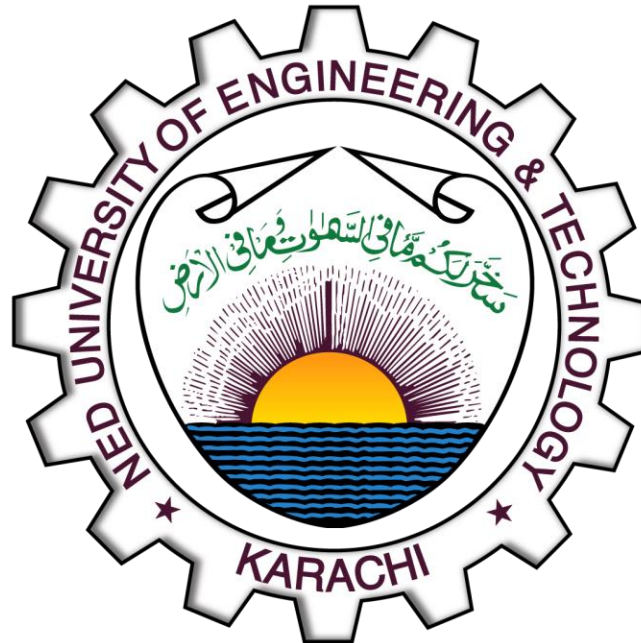


OBJECT ORIENTED PROGRAMMING

[CT-260]



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LAB :#

QUESTION#1:

CODE:

```
#include <iostream>
#include <fstream>
using namespace std;

int main()
{
    fstream fout("./data.txt", ios::out | ios::trunc);
    if (!fout)
    {
        cout << "An error occurred." << endl;
        return 1;
    }
    string input;
    cout << "Input data you want to store: ";
    getline(cin, input);
    cout << "Length: " << input.length() << endl;
    for (char c : input)
        fout.put(c);
    fout.close();
    ifstream fin("./data.txt");
    char ch;
    cout << endl << "\e[1;32mData retrived from the file.\e[33m" << endl;
    while (fin.get(ch))
        cout << ch;
    cout << "\e[0m\n";
    return 0;
}
```

OUTPUT:

```
Input data you want to store: Hello, this is some data that will be saved on a  
text file
```

```
Length: 58
```

```
Data retrived from the file.
```

```
Hello, this is some data that will be saved on a text file
```

Text file:



OOPspring26 [WSL: Ubuntu-24.04] - data.txt

```
1 Hello, this is some data that will be saved on a text file
```

QUESTION#2:

CODE:

```
#include <iostream>
#include <fstream>

using namespace std;

int main()
{
    ifstream sourceFile("input.txt", ios::in);
    if (!sourceFile)
    {
        cout << "Error: Cannot open the source file 'input.txt!'" << endl;
        return 1;
    }

    ofstream destFile("copy_of_input.txt", ios::out);
    if (!destFile)
    {
        cout << "Error: Cannot create or open the destination file!" << endl;
        sourceFile.close();
        return 1;
    }

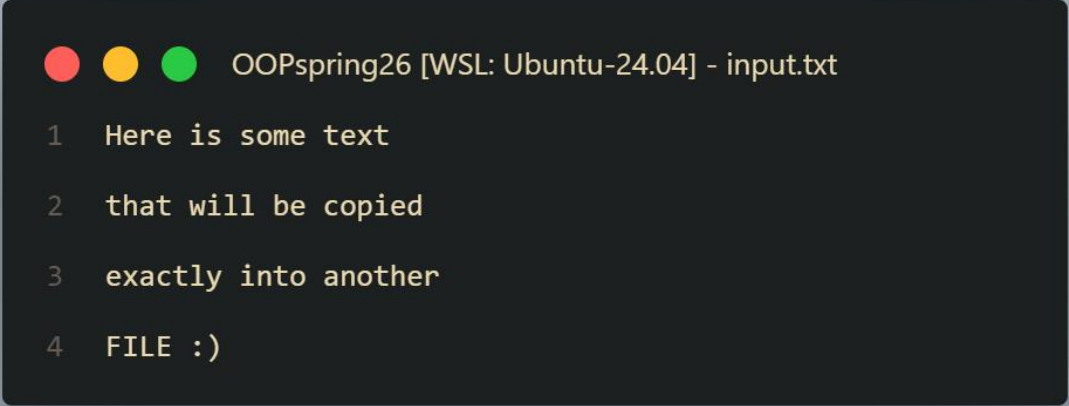
    char ch;
    while (sourceFile.get(ch))
        destFile.put(ch);

    cout << "\e[1;32mFile copied successfully in text mode!\e[0m" << endl;
    sourceFile.close();
    destFile.close();
    return 0;
}
```

OUTPUT:

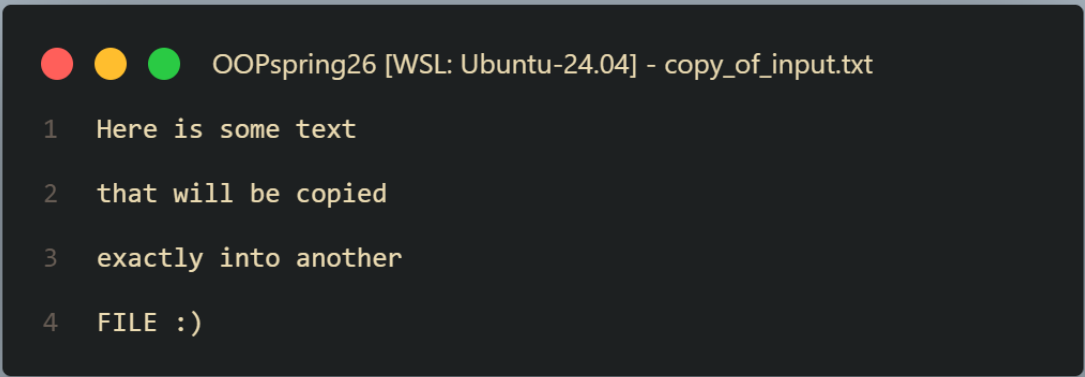
File copied successfully in text mode!

INPUT FILE:

A terminal window with a dark background and light blue title bar. The title bar text is "OOPspring26 [WSL: Ubuntu-24.04] - input.txt". The terminal content shows four lines of text, each preceded by a line number from 1 to 4.

```
OOPspring26 [WSL: Ubuntu-24.04] - input.txt
1  Here is some text
2  that will be copied
3  exactly into another
4  FILE :)
```

COPIED FILE:

A terminal window with a dark background and light blue title bar. The title bar text is "OOPspring26 [WSL: Ubuntu-24.04] - copy_of_input.txt". The terminal content shows four lines of text, each preceded by a line number from 1 to 4, identical to the input file.

```
OOPspring26 [WSL: Ubuntu-24.04] - copy_of_input.txt
1  Here is some text
2  that will be copied
3  exactly into another
4  FILE :)
```

QUESTION#3:

CODE:

```
#include <iostream>
#include <fstream>
#include <cstring>
using namespace std;

class Person
{
private:
    char name[25];
    int age;

public:
    Person() {}
    Person(char *name, int age) : age(age)
    {
        strncpy(this->name, name, 24);
        this->name[24] = '\0';
    }
    void print()
    {
        cout << name << endl
              << age << endl;
    }
};

int main()
{
    char name[25];
    int age;
    cout << "Enter your name: ";
    cin.getline(name, 24);
    cout << "Enter your age: ";
    cin >> age;
    Person person1(name, age);
    fstream file("./person.bin", ios::binary | ios::out | ios::trunc);
    if (!file)
    {
        cout << "\e[1;31mFile can't be generated.\e[0m\n";
        return 1;
    }
}
```

```

file.write((char *)&person1, sizeof(person1));
file.close();

file.open("./person.bin", ios::binary | ios::in);
if (!file)
{
    cout << "\e[1;31mFile not available.\e[0m\n";
    return 1;
}
Person person2;
if (file.read((char *)&person2, sizeof(person2)))
{
    cout << "-----\e[1;32mInfo retrived from the file\e[0m-----\n";
    person2.print();
}
else
    cout << "\e[1;31mAn error occured while reading the file.\e[0m\n";
return 0;
}

```

OUTPUT:

```

Enter your name: Taha Ahmed
Enter your age: 18
-----Info retrived from the file-----
Taha Ahmed
18

```

FILE (./person.bin):

char name[25]

padding

int age=18

Hex	Decoded Text
00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F	Taha Ahmed
00 00 00 00 00 00 00 00 00 00 43 A7 94 12 00 00 00	...C...

QUESTION#4:

CODE:

```
#include <cstring>
#include <iostream>
#include <fstream>
using namespace std;

class Participant
{
private:
    int ID, score;
    char name[25];

public:
    Participant() : ID(0), score(0)
    {
        name[0] = '\0';
    }
    Participant(int ID, char *name, int score) : ID(ID), score(score)
    {
        strncpy(this->name, name, 24);
        name[24] = '\0';
    }

    int getID() { return ID; }
    int getScore() { return score; }
    void print()
    {
        cout << "Name: " << name << endl;
        cout << "ID: " << ID << endl;
        cout << "Score: " << score << endl;
    }

    void input()
    {
        fstream file("./participant.dat", ios::app | ios::binary);
        if (!file)
        {
            cout << "\e[1;31mFile error.\e[0m\n";
            return;
        }
        file.write((char *)this, sizeof(Participant));
    }
};
```



```

}

void output(int IDin)
{
    Participant p;
    fstream file("./participant.dat", ios::in);
    if (!file)
    {
        cout << "\e[1;31mNo Participants.\e[0m\n";
        return;
    }
    while (file.read((char *)&p, sizeof(p)))
    {
        if (p.getID() == IDin)
        {
            cout << "-----\e[1;32mTarget Found\e[0m-----\n";
            p.print();
            return;
        }
    }
    cout << "\e[1;31mID wasn't in the file.\e[0m" << endl;
}

void max()
{
    Participant p, max;
    fstream file("./participant.dat", ios::in | ios::binary);
    if (!file)
    {
        cout << "\e[1;31mNo Participants.\e[0m\n";
        return;
    }
    while (file.read((char *)&p, sizeof(p)))
        if (p.getScore() > max.getScore())
            max = p;
    cout << endl
        << "\e[1;32mMaxima:\e[0m" << endl;
    max.print();
}
};

int main()
{
    int ID, score;
    char name[25];

```

```
cout << "Enter your name: ";
cin.getline(name, 24);
cout << "Enter ID: ";
cin >> ID;
cout << "Enter score: ";
cin >> score;

Participant p(ID, name, score);

p.input();

cout << "Enter ID you want to search: ";
cin >> ID;
p.output(ID);

p.max();

return 0;
}
```

OUTPUT:

```
Enter your name: Danial
Enter ID: 2
Enter score: 130
Enter ID you want to search: 1
-----Target Found-----
Name: Taha
ID: 1
Score: 150

Maxima:
Name: Taha
ID: 1
Score: 150
```

FILE (./participants.dat):

The screenshot shows a hex editor window titled 'participant.dat M'. The main area displays hex data with corresponding decoded text on the right. Handwritten annotations identify key fields:

- ID:** Points to the hex value `02` at offset `00000004`.
- score:** Points to the hex value `96` at offset `00000005`.
- name[25]:** Points to the decoded text `Taha` starting at offset `0000000A`.
- Padding:** Points to the initial zero-filled bytes at offset `00000000`.

Offset	Hex	Decoded Text
00000000	01 00 00 00 96 00 00 00 54 61 68 61 00 00 00 00	Taha.....
00000010	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
00000020	67 6C 69 62 02 00 00 00 82 00 00 00 44 61 6E 69	glib.....Danial
00000030	61 6C 00 00 00 00 00 00 00 00 00 00 00 00 00 00	al.....
00000040	00 00 00 00 67 6C 69 62 +	glib+

QUESTION#5:

CODE:

```
#include <iostream>
#include <fstream>
using namespace std;

int main()
{
    fstream file("./STORY.txt", ios::in);
    if (!file)
    {
        cout << "\e[1;31mFile not found.\e[0m\n";
        return 1;
    }
    string line;
    uint count = 0;
    while (getline(file, line))
        if (line[0] != 'A')
            count++;
    cout << "\e[1;32mTotal: " << count << "\e[0m\n";
    return 0;
}
```

OUTPUT:

```
Total: 3
```

FILE (./STORY.txt):

```
● ● ● OOPspring26 [WSL: Ubuntu-24.04] - STORY.txt

1 The rose is red.
2 A girl is playing there.
3 There is a playground.
4 An airplane is in the sky.
5 Numbers are not allowed in the password.
```